A.3. Deep learning

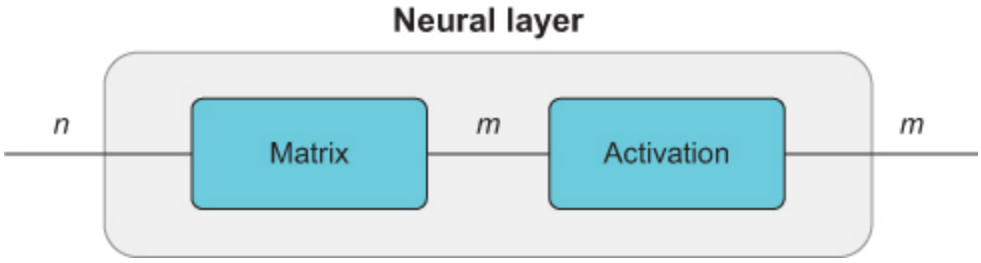
A deep neural network is simply a composition of multiple layers of simpler functions called *layers*. Each layer function consists of a matrix multiplication followed by a nonlinear *activation function*. The most common activation function is $f(x) = \max(0,x)$, which returns 0 if x is negative or returns x otherwise.

A simple neural network might be



Read this diagram from left to right as if data flows in from the left into the L1 function then the L2 function and becomes the output on the right. The symbols *k*, *m*, and *n* refer to the dimensionality of the vectors. A *k*-length vector is input to function L1, which produces an *m*-length vector that then gets passed to L2, which finally produces an *n*-dimensional vector.

Now let's look at what each of these L functions are doing.



A neural network layer, generically, consists of two parts: a matrix multiplication and an activation function. An *n*-length vector comes in from the left and gets multiplied by a matrix (often called a parameter or weight matrix), which may change the dimensionality of the resulting output vector. The output vector, now of length *m*, gets passed through a nonlinear activation function, which does not change the dimensionality of the vector.